EE5630 Digital Signal Processing

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Homework #4: discrete-time approximation of continuous-time systems

Due Friday Nov. 5, 2021

1. [50 pts] (**Analysis of a simplified model for the car suspension system**). Figure 1 depicts a one-dimensional model for a car-suspension system. Assume that the car is going toward $+y$ direction at a constant speed v. Also assume that the deformation of the tire is negligible, so the vertical displacement of the wheel x_1 directly reflects the road condition. We then write $x_1 = g(y)$. The physical parameters are defined and given as follows:
   - k = 28,800 Nt/m (elastic constant),
   - b = 1500 kg/s (damping coefficient), and
   - m = 800 kg (mass).

   We are interested in calculating the vertical displacement $x_2(t)$ of the automobile body in response to $x_1(t)$.

   (a) Assume that the car has been going through a smooth ideal road until time $t = 0$, so we can set the initial conditions $x_2(0) = x_1(0) = 0,$ and $\dot{x}_2(t) = 0$ at $t = 0$. Based on basic laws of Newton's mechanics, show that the transfer function is given as follows:

   $$H(s) = \frac{X_2(s)}{X_1(s)} = \frac{bs + k}{ms^2 + bs + k}$$

   (b) Plot the magnitude spectrum $|H(j2\pi f)|$ as a function of frequency f. At what frequency approximately does it reach the maximum? Interpret your results. [MATLAB function calls: abs( ), loglog(), semilogx(), or semilogy()]

   (c) Plot the group delay $\tau_g = -\partial\measuredangle H(j\Omega)/\partial\Omega$ as a function of frequency [MATLAB function calls: angle(), diff()].

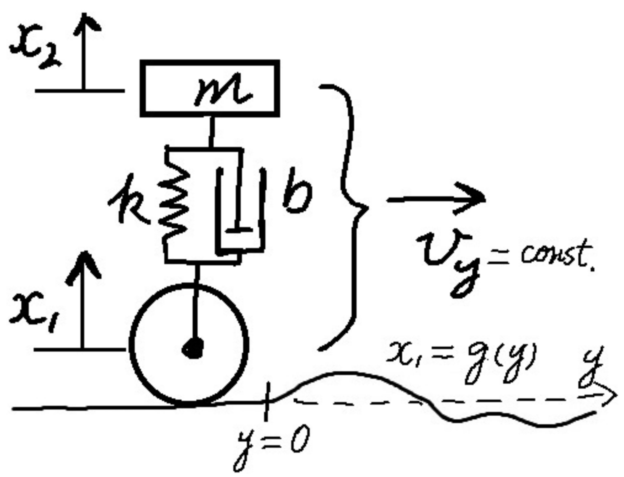

Fig. 1: A one-dimensional model for the car suspension system.

2. [50 pts] (Continuing from 1, **discrete-time modeling**). Now let us simulate the system's response to a periodic series of bumps shown in Fig. 2. The road condition can be described approximately as

$$g(y) = D\sin\left(2\pi\frac{y}{L}\right), 0 < y < \frac{L}{2}$$

$$g(y) = 0,\ \frac{L}{2} < y < L, \text{ and}$$

$$g(y+L) = g(y), \text{ for all } y > 0.$$

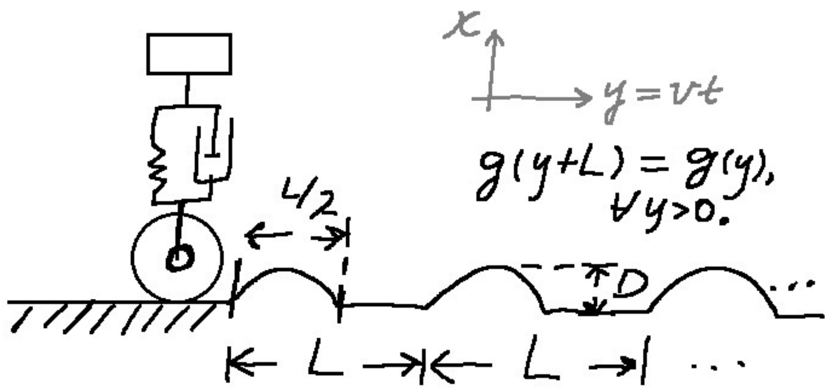

Fig. 2: A series of bumps that can be described as a half-wave rectified sine function

(a) **Finite-difference schemes**. Approximate the system equation by replacing $\dot{x}_2(t)$ with $(x_2[n+1] - x_2[n-1])/2T$ and $\ddot{x}_2(t)$ with $(x_2[n+1] - 2x_2[n] + x_2[n-1])/T^2$. Do the approximation similarly for $x_1(t)$. Derive a difference equation that specifies the relation between $x_2[n]$ and $x_1[n]$.

(b) Assume that the car goes toward the right at v = 10 m/s, the distance between bumps is L = 8 m, and the height of the bumps is D = 5 cm. <u>Choose a sufficiently short period T</u> and sample the input signal as follows:

$$x_1[n] = g(y)|_{y=vnT}$$

Derive the system function $H(z) = X_2(z)/X_1(z)$. Then, calculate the displacement $x_2[n]$ in response to the road condition $x_1[n]$ for time 0-10 s. You may use MATLAB's filter() function. Plot the response and find out the maximum displacement.

(c) Is the maximum displacement greater than or smaller than D? What if the velocity is 20 m/s or 5 m/s instead?

3. Bonus [10 pts] (Continuing from Problem 1, **discretization by sampling the impulse response**).
   (a) Write down the impulse response of the system by finding the inverse Laplace transform of the transfer function $H(s)$.
   (b) Sample the impulse response $h(t)$ at a sufficiently high sampling rate f_s. You may truncate the result at a sufficiently late time, i.e., after $h(t)$ decays to a low level. Use MATLAB's freqz() to calculate the magnitude and phase response of the approximate system from frequency 0 to $f_s/2$. Compare the results with results you obtained in 1(b) and 1(c).

For this homework, please turn in your code and a brief report.